



Glyphosate-induced stiffening of HaCaT keratinocytes, a Peak Force Tapping study on living cells

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ABSTRACT

The skin is the first physiological barrier, with a complex constitution, that provides defensive functions against multiple physical and chemical aggressions. Glyphosate is an extensively used herbicide that has been shown to increase the risk of cancer. Moreover there is increasing evidence suggesting that the mechanical phenotype plays an important role in malignant transformation. Atomic force microscopy (AFM) has emerged within the last decade as a powerful tool for providing a nanometer-scale resolution imaging of biological samples. Peak Force Tapping (PFT) is a newly released AFM-based investigation technique allowing extraction of chemical and mechanical properties from a wide range of samples at a relatively high speed and a high resolution. The present work uses the PFT technology to investigate HaCaT keratinocytes, a human epidermal cell line, and offers an original approach to study chemically-induced changes in the cellular mechanical properties under near-physiological conditions. These experiments indicate glyphosate induces cell membrane stiffening, and the appearance of cytoskeleton structures at a subcellular level, for low cytotoxic concentrations whereas cells exposed to IC50 (inhibitory concentration 50%) treatment exhibit control-like mechanical behavior despite obvious membrane damages. Quercetin, a well-known antioxidant, reverses the glyphosate-induced mechanical phenotype.

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1. Introduction

The skin is the first physiological barrier, with a complex constitution, that provides defensive functions against multiple physical and chemical aggressions (Proksch et al., 2008). Its three-layer composition (epidermis, dermis and hypodermis) imparts very particular mechanical properties. The outermost layer, the epidermis, is made of four stratum of keratinocytes, and these cells are continuously exposed to environmental pollutants including herbicides.

Epidemiological studies (Bolognesi et al., 2009; De Roos et al., 2003) have shown that chronic exposure to glyphosate, an extensively used herbicide, increases the risk of cancer. More recently, it has been shown that glyphosate-based formulations had teratogenic effects on *Xenopus laevis* embryos (Paganelli et al., 2010), and

that glyphosate alone induced clastogenic effects (Prasad et al., 2009) and carcinogenicity (George et al., 2010) respectively in murine bone marrow and skin. Moreover, two recent reviews highlighted the importance of cell mechanical phenotype in malignant transformation (Kumar and Weaver, 2009; Suresh, 2007).

On an experimental basis, glyphosate is a chemical component that is both studied alone and in formulations for its deleterious effects on different cell types (Benachour et al., 2007; Marc et al., 2002). Although some publications have reported cases of cutaneous toxicity with glyphosate or its formulations (Amerio et al., 2004; Nagami et al., 2005; Penagos et al., 2004), little is known about the morphological and mechanical consequences of its toxicity towards the human skin and its constitutive cells (Elie-Caille et al., 2010).

Atomic force microscopy (AFM) has emerged within the last decade as a powerful tool for various physical and biological applications; it has evolved from an imaging technique to a multifunctional “lab on a tip”. Indeed, AFM based force spectroscopy is increasingly used to study the mechanisms of molecular recognition and protein folding/unfolding, to probe chemical groups and dynamics of receptor–ligand interactions (Florin et al. 1994; Dorobantu and Gray, 2010; Rico et al., 2011), and to study the local

Abbreviations: AFM, atomic force microscopy; PFQNM, Peak Force Quantitative Nanomechanical Mapping; PFT, Peak Force Tapping; YM, Young's modulus.

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elasticity (Clausen-Schaumann et al., 2000) and the mechanical properties of soft biological samples (Butt et al., 2005; Müller and Dufrene, 2008). The AFM has also provided nanometer-scale resolution imaging of biological samples ranging from single molecules, such as DNA (Hamon et al., 2007), to intact cells attached on biomaterials (Berquand et al., 2010). Emerging applications include increasing the resolution of mechanical measurements in biological contexts such as cell division (Gilbert et al., 2007; Stewart et al., 2011) or cell adhesion (Li et al., 2003; Fierro et al., 2008).

Contact mode AFM (Binnig et al., 1986) is often referred to as the preferred technique to image soft biological for its ease-of-use. Tapping mode (Zhong et al., 1993) also has advantages since the tip only intermittently contacts the surface thereby allowing for negligible friction and shear forces on the sample. Associated with tapping mode is the phase signal, which is defined as the lag between the motion of the cantilever's drive and its response, and it reflects the energy, dissipated between the tip and the sample during each tap on the surface. While the phase signal contains information about the sample's properties, it suffers from the significant drawback of being physically ambiguous. As noted, the phase signal is composed of any tip-sample energy dissipation mechanism, so factors such as adhesion, viscosity, elasticity are all inextricably intertwined, with no possibility to obtain the contributions from each of the individual factors separately.

Peak Force Tapping (PFT) (Pittenger et al., 2010) is an AFM-based operational mode that enables the extraction of chemical and mechanical properties from a wide range of samples at a relatively high speed (similar to Tapping Mode) and a high resolution. If the probe is calibrated onto a known sample prior to the experiment, quantitative nano-mechanical property information can be directly extracted from the sample. In this technique, the z piezo is driven with a sinusoidal waveform (not triangular like in regular force–distance curves) at a drive frequency far below the cantilever resonance frequency (typically 1–2 kHz depending on the instrument) and the drive amplitude is usually 150 nm (300 nm peak-to-peak). Unlike in tapping mode where the feedback is based on the amplitude value, the imaging feedback is based on the Peak Force of the force–distance curves. Fig. 1 (left) shows how the tip is subject to negative (attractive) and positive (repulsive) forces during an approach–retract cycle. Fig. 1 (right) represents how the different mechanical properties can be extracted from each force curve. Since this process is repeated for each pixel of the image, the resolution of all of the mechanical channels will be the same as that of a conventional height AFM image. Once the system is calibrated the Peak Force error, the Young's modulus (YM), the dissipation, the deformation and the adhesion can be directly displayed in the quantitative units of m, Pa, eV, m and N, respectively.

Hence, this PFT technique offers the possibility of mapping the entire cellular mechanical properties at high resolution.

Recent nanomechanical studies on bone cancer cells have revealed heterogeneity in the collagen fibrils and have been used to propose new techniques for early detection of osteoarthritis (Stolz et al., 2009). In the case of metastatic cancer cells, cell elasticity measurements revealed more than 70% decreases in the cancerous cell stiffness compared to the benign cells that line the body cavity (Cross et al., 2007). Additionally, a nanoscale mapping of yeasts has revealed areas on the cell surface being significantly stiffer than the surrounding cell wall (Touhami et al., 2003).

Among the limited number of AFM studies recently conducted with cellular models, few have been proposed on living mammalian cells (Fung et al., 2010). Within these cases, measurements of mechanical properties are mostly performed on a limited number of points (Gunning et al., 2008). Restricting these measurements to only small portions of the cell does not represent the entire cell behavior and may lead to invalid characterization. Our PFT approach could help characterizing cells with a much higher accuracy since it collects high resolution data on the whole cell surface. Up to now, this very promising technique has only been used to investigate mechanical properties of α -amyloid fibrils (Adamcik et al., 2011), fixed cells (Berquand et al., 2011) or living diatoms (Pletikapic et al., *in press*) but has never been thoroughly tested on human eukaryotic cells.

The present work uses the PFT technology to study a human epidermal cell line, HaCaT, and offers an original approach to track chemically-induced changes in cell mechanical properties under near-physiological conditions.

2. Materials and methods

2.1. Cell cultures

The immortalized human HaCaT cell line (ATCC, Teddington, UK) (Boukamp et al., 1988) was cultured in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% (v/v) fetal calf serum (FCS; D. Dutscher, Brumath, France) and 80 mg/L of gentamicin in a humidified 5% CO₂ atmosphere at 37 °C. The keratinocytes were grown to confluence in 75 cm² culture flasks (D. Dutscher). The medium was removed every 48 h, and cells were subcultured every 7 days at a ratio of 1:6.

The *in vitro* conditions to study the glyphosate-induced stress were determined by carrying out cytotoxicity assays on 96-well microtiter plate HaCaT cultures (Elie-Caille et al., 2010) with different glyphosate concentrations. The cytotoxic profiles remained unchanged from 6 to 18 h of contact with glyphosate, showing a

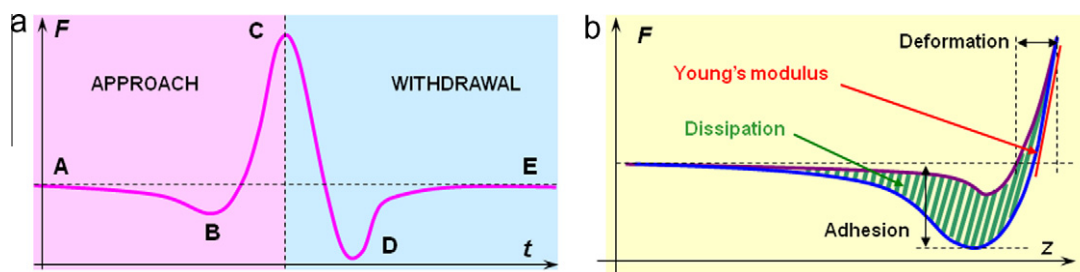


Fig. 1. PFQNM guiding principles (a) graph of the AFM tip force interactions during a single approach–retract cycle. A: the tip is far off the surface and experiences an attractive field first until it reaches the contact point (B), after which the tip experiences a repulsive field as it begins to indent into the sample. C represents the maximum Peak Force that is used for the feedback (or to trigger the retraction portion of the cycle). In the second part of the cycle, the tip starts pulling off until it reaches its maximum adhesion (D) and then returns to its starting position E & A. (b) plots the force vs. z -displacement and explains how the different material properties are extracted from the force curve. The adhesion is calculated the same as in regular force mode and is the vertical distance between the base line and the bottom most part of the retraction curve. The deformation is the horizontal distance between the contact point and the turn-away point (maximum indentation). The Young's modulus can be extracted by extrapolating the linear portion of the retraction curve after the contact point. The dissipation can be calculated by integrating the area between the two curves.

similar 50% inhibitory concentration (IC₅₀) value, around 30 mM. We chose to test 30 mM to observe achieved events and 15 mM (80% cytotoxicity) for ongoing events.

2.2. AFM

In AFM studies, HaCaT cells were seeded in 50 mm glass bottom Petri dishes (Willco Wells BV, Amsterdam, The Netherlands) at a density of 5×10^5 cells in 3 mL DMEM supplemented with 10% (v/v) FCS and 80 mg/L of gentamicin in a humidified 5% CO₂ atmosphere at 37 °C for 24 h, for cell attachment. Cultured cells were exposed to several concentrations of glyphosate in FCS-free media depending on different incubation periods: 40 and 53 for 0.5 h; 15 and 30 for 6 and 18 h. A 100 mM quercetin treatment was tested concomitantly to 18 h glyphosate conditions. Following exposure the cells were washed with pre-warmed FCS-free media and imaged at ambient temperature which was found to be 32 °C and over a maximum time of 90 min to be sure that the cells were still healthy and adherent during the imaging. AFM imaging was carried out on a Bioscope Catalyst™ (Bruker, Billerica, USA). AFM probes used in Peak Force Quantitative Nanomechanical Mapping (PFQNM) mode were ScanAsyst-Fluid (Bruker, Billerica, USA) probes. The Young's moduli were calculated by using a Sneddon fit (Sneddon, 1965).

For each sample, a minimum of four moderate resolution (128×128) images and one high resolution (512×512) image were captured. The probe's spring constant was found to be 0.42 N/m. For the Sneddon fit, we used an opening angle of 20° and a Poisson's ratio of 0.4 based on previous work (Trickey et al., 2006). The average loading force applied to the sample was 632 pN which is significantly below the typical value of 1 nN representative of the force applied to biological samples in tapping mode or contact mode.

Prior to the experiment and in order to express directly quantitative data, the AFM probe has been calibrated as follows: first, the tip is engaged on a stiff sample (glass), a representative force curve is captured, and the deflection sensitivity calculated. Second, the tip is withdrawn and the spring constant is calculated by using a thermal tune (a sweep in frequency allows finding the resonance peak). Finally, the tip radius is automatically calculated by scanning a "tipcheck" sample, which enables the calculation of the tip shape by using the blind reconstruction technique provided in the Nanoscope Analysis software. This last step is of particular importance since the sample's mechanical properties, especially the Young's modulus, are strongly dependent on the tip shape. Calibration can also be accomplished by imaging a sample having known mechanical properties with the same tip that will be used during the experiments.

2.3. Tensile modulus calculation and fit model

The basic principle of PFT, as well as the different information which can be extracted, are detailed in Fig. 1. The YM or tensile modulus is obtained by dividing the tensile stress by the tensile strain and is related to the stiffness of the scanned material. The stiffer the material, the higher the YM. With respect to the YM calculation, many publications refer to the Hertz model (Hertz, 1881) which has the advantage of simplicity in that ignores the adhesion force between the tip and the sample. This is also a significant drawback of the model; however it is nonetheless used, particularly when working in fluid where adhesion forces are minimized. In our experiments, two different models, both which extend the Hertz model, were applied.

By default, the AFM's software (Nanoscope V8.2) uses a DMT fit (Derjaguin et al., 1975) to extract the Young's modulus from each force curve. The DMT model assumes that the adhesion forces

are mainly focused in a ring located at the edge of the contact area. The images were processed by using the "bearing analysis" (Bruker Analysis Software). The obtained value is an average of 300,000 force curves. While this model is suitable with non- (or slightly) compliant samples like high-density polymers, it is not particularly suitable for biological samples.

Thus, secondly, the data was also processed outside of the AFM software using the Sneddon model (Sneddon, 1965). Sneddon takes into consideration the deformation induced by the indenter and is more commonly used on biological samples (Domke and Radmacher, 1998; Rotsch and Radmacher, 2000; Berdyeva et al., 2005). Because this model is not native to the AFM, for each image, a high speed data capture (HSDC) (5400 points) of the force curve signal was obtained. Creating multiple HSDC files, enabled the raw force curves to be exported and post-processed by using a custom program, Rainbow (Belikov et al., 2009), which gives the user the possibility to apply several different models, including Sneddon, to extract Young's modulus. The force curves are captured on a representative part of the image where only cells are present rather than the scaffold. Although, only one probe was used for the entire series of experiments, the Young's modulus calculated from the exported force curves is an average of 27,000 data points and Young's modulus values extracted using a DMT fit are superimposed to those of another series of experiments carried out in the same conditions.

3. Results

3.1. Force images

In control conditions (Fig. 2ai), cells are spread and interconnected by adhesion membrane expansions. The membrane surfaces show native protrusions characterizing control cells (Reich et al., 2007, 2009). Though the cellular surface can easily be tracked, the features observed on the resulting topography-related images like the Peak Force error channel seem to indicate that the top most part of the scanned cells is rather fluid and mobile, which might mean that the overall cells have a certain degree of freedom and motility. This is confirmed by the fact that the cells have slightly moved after several scans at the same location (data not shown here). The YM representation (Fig. 2aai) shows a homogeneous cell surface, except in the intercellular spaces where the YM increases greatly. It is speculated that this is due to a thinning, and in that case, the influence of the scaffold on the YM becomes non negligible. This is particularly apparent when contrasting cells are distant from those in close-contact (Fig. 2aai, asterisk). As expected, the depth of the deformation is higher for the softest portions of the sample and vice versa. As an example, the glass (bright yellow in Fig. 2aiii) does not deform whereas the core part of the cell (darker) is much softer.

After a gentle cytotoxic treatment (6 h, 15 mM-glyphosate), the topography profile changes (Fig. 2bi). The cells exhibit a flattened membrane and a different distribution of native protrusions. They also show a complex subcellular filamentous network with numerous membrane junction points. The YM values (Fig. 2bii) are very high (bright contrast) for these particular structures. As expected, those features correspond to very small deformations (Fig. 2biii).

3.2. Topography data

Under different treatment conditions, topography images (Fig. 3) give various glyphosate-induced cellular morphology profiles.

After 6 and 18 h-incubation, control cells show similar membrane surfaces with regular and homogenous protrusions. Cells are spread and membrane expansions are clearly visible demon-

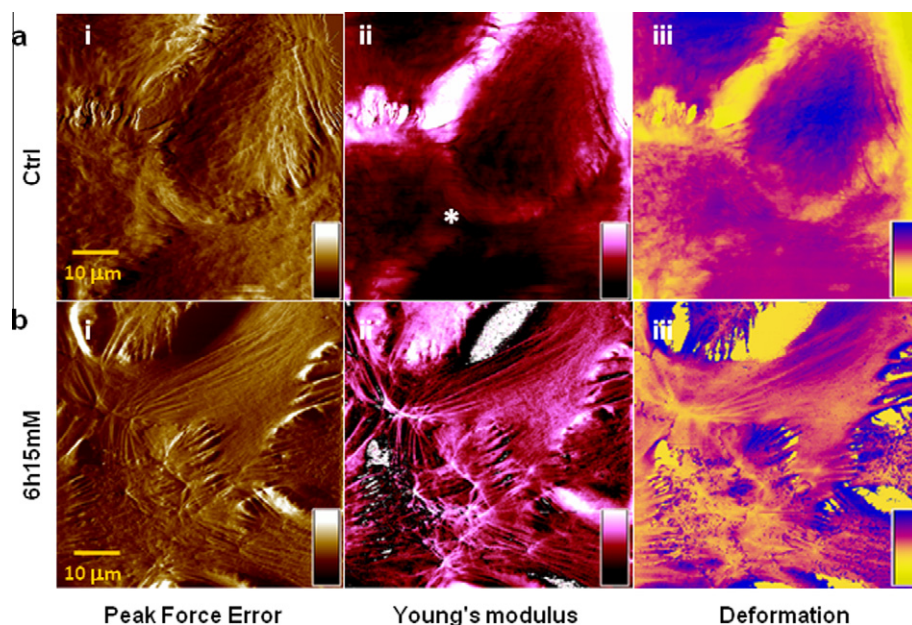


Fig.2. AFM images of 6 h–15 mM glyphosate-treated cells vs. untreated control HaCaT cells characterizing topography (Peak Force Error channel, which is the derivative of topography in the scan direction (i)) and mechanical measurements (Young's modulus (ii) processed in real time by using a DMT fit, deformation (iii)) for control cells (a) and treated cells (b) at 15 mM glyphosate for a 6 h–incubation. (*) shows an area of cell–cell close contact. The color scales are (from bottom to top): 0–1 nN, 0–500 kPa and 0–250 nm for the Peak Force Error, Young's modulus and deformation images, respectively.

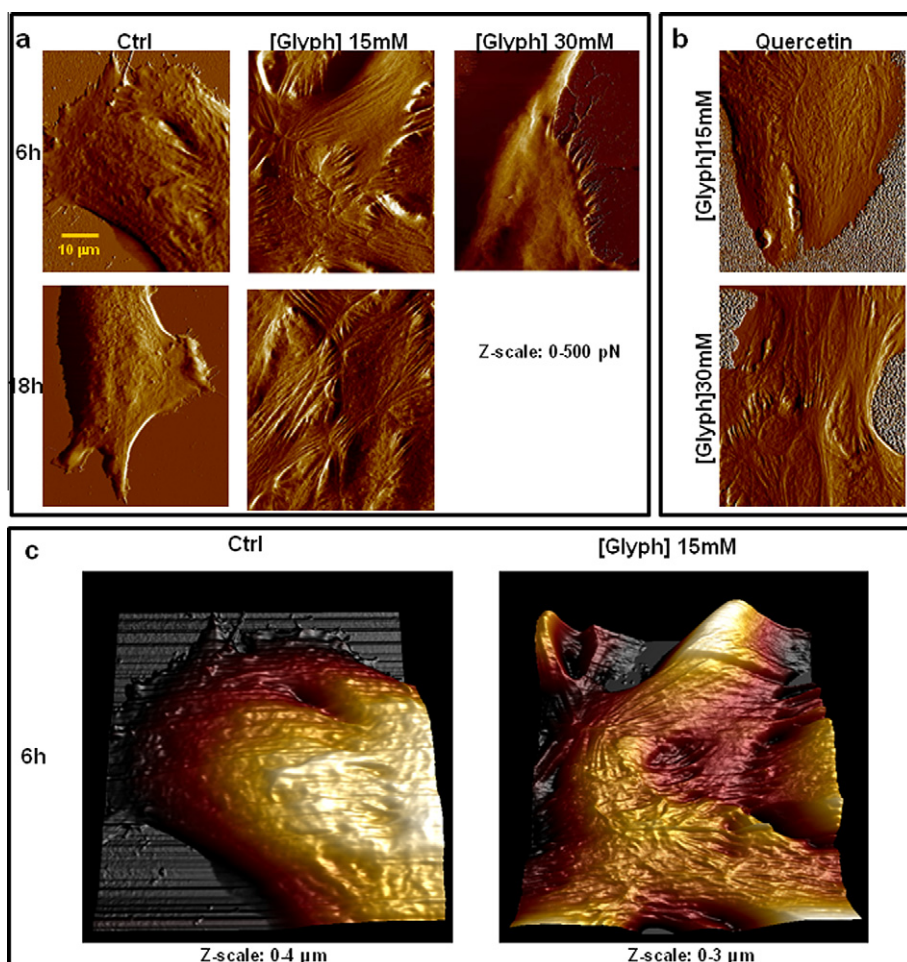


Fig.3. Chemically induced HaCaT cell morphological impairments (a) alteration of HaCaT cell integrity induced by glyphosate after 6 h and 18 h-incubation time; (b) the reversing effect of a quercetin treatment (100 mM).(c) Highlights of 3D corresponding images for 6 h untreated control cells ($z_{\max} = 4 \mu\text{m}$) and glyphosate 15 mM treated cells ($z_{\max} = 2.5 \mu\text{m}$).

strating that the cells tend to expand and reach out to one another. Nuclei form subcellular globular structures in the middle of the cells.

After a 15 mM-glyphosate treatment, cells develop a filamentous subcellular network. These filaments seem to be linked together by membrane junctions. The cellular volume distribution is significantly different from the control and can be better observed in the corresponding 3D images (Fig. 3c). Cells are less delimited, closely packed together and the nuclei are no longer visible at the subcellular scale.

After an IC50 treatment (30 mM-glyphosate for 6 h), cells are flattened and show weak reticulations. At 30 mM-glyphosate for 18 h, only low resolution images were carried out (data not shown). In this particular condition, cells exhibit the same profile.

To further these results, we tested the potential of reversion of a quercetin treatment on the glyphosate-induced morphological changes as described above (Fig. 3b).

After a concomitant 100 mM-quercetin/15 mM-glyphosate treatment, cells show protrusions and a weak membrane network which are different from the observations in 15 mM-glyphosate conditions: it seems there is an intermediate profile between control and 15 mM-glyphosate conditions.

A 100 mM-quercetin/30 mM-glyphosate exposure reveals a cellular profile comparable to that of the treatment with 15 mM-glyphosate only, i.e. with the presence of a filamentous network.

3.3. Effect of glyphosate +/- quercetin on YM and deformation parameters

The control cells show YM values respectively amounting to about 77 and 109 kPa for 6 and 18 h incubations (Fig. 4a). After a 15 mM-glyphosate treatment, these values increase approximately 4-fold for 6 h and 3-fold for 18 h. These increases reflect a significant rise in cell stiffness. Conversely, in the case of a 30 mM-glyphosate treatment, YM values are similar to those of the cell control (YM is about 104 and 125 kPa for respectively 6 and 18 h incubation).

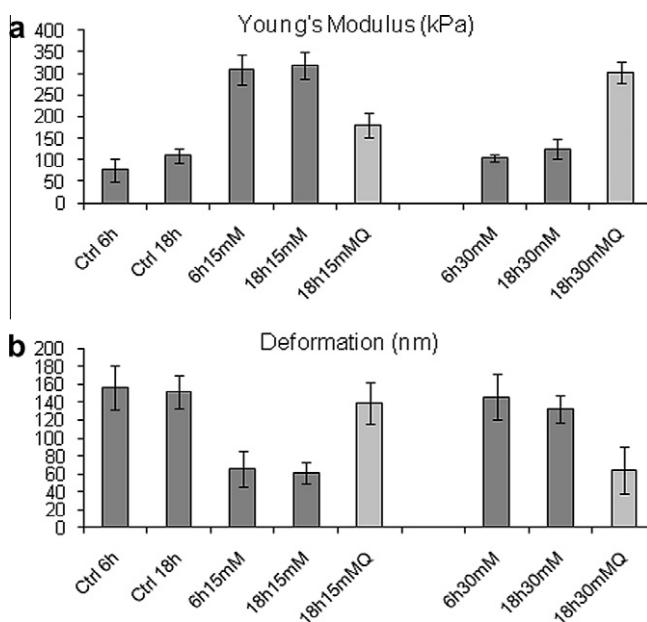


Fig. 4. Statistical analysis of Young's modulus (calculated by using a Sneddon fit) (a) and deformation (b). Statistics were drawn up for five different images for control, glyphosate treated and quercetin/glyphosate treated cells.

The quercetin supplementation concomitant to the glyphosate treatment leads to a change in cell stiffness: after a 15 mM-glyphosate treatment, quercetin causes the average YM to decrease by factor 1.8 (319 to 181 kPa). On the contrary, after a 30 mM-glyphosate treatment, quercetin causes the average YM to increase by factor 3 (125 to 301 kPa).

YM and deformation are complementary measurements: in Fig. 4b, deformation and YM variations are opposed to each other. This observation reinforces the idea that glyphosate treatment induces important modifications of keratinocyte mechanical properties.

4. Discussion

The skin is the largest organ of the human body and the first barrier against environmental pollutants including herbicides. Alone or as the active component of commercial formulations, glyphosate has long been considered as harmless in normal use and in chronic exposure testing (Williams et al. 2000). However, it has also been demonstrated that glyphosate is able to go through the human skin, where it predominately remains in the epidermis; the glyphosate penetration in healthy skin is 5-fold better than in compromised skin (Nielsen et al., 2007).

Living cells are complex and heterogeneous viscoelastic structures; their mechanical properties are known to be closely related to the health and function of the human body. The onset and progression of some disease states can be strongly associated with the mechanical properties of living cells (Chouinard et al., 2008; Hsieh et al., 2008; Suresh et al., 2005). There are a variety of aging theories in the literature, but a common underlying factor is the structural change at several scales, be it the thinning of the epidermis, or the reduction of skin strength and elasticity. Moreover, several studies have demonstrated the correlation between aging and cellular stiffening, more particularly due to cytoskeleton "shake-up" (Lieber et al., 2004; Starodubtseva, 2011). There is also increasing evidence that the mechanical properties of keratinocytes also participate in skin aging mechanisms.

AFM has proven to be a useful tool in ultrastructural biology as it is capable of providing a quantitative description of the morphological details, and physical properties, under physiological conditions. In our study, we were able to provide unique high resolution force measurement images and statistical data through the application of PFQNM. As a mechanical fit, we have used a custom implementation of the Sneddon model, as opposed to the DMT model provided by the instrument. For biological applications (in particular soft samples like living cells) this gives a better approximation of the elasticity contact mechanics.

Systematic comparison of high resolution topography images and force data provides a good outline of the cellular morphological state. The mechanical characteristics are as expected for control conditions – the intercellular spaces are stiffer than the rest of the cell (which is more voluminous). Laurent et al. (2005) have characterized keratinocytes lamellipodia elasticity exhibiting a tendency to decrease from ~55 kPa at the front of the lamellipodium to ~10 kPa at the rear. Moreover, the YM values are in the range of common living cells (Alonso and Goldmann, 2003). The filamentous network appears at a subcellular level after a 15 mM-glyphosate treatment. These structures have a high YM value and appear to be cytoskeleton.

The comparison of several topography images, and their statistics, shows similar morphological profiles with increasing stiffness for a 15 mM-glyphosate treatment after 6 and 18 h-incubation period. While it is unclear of the specific mechanism, the stiffness increase can be attributed to changes in any of the three different types of cytoskeleton structures: actin, keratin and tubulin.

Considering actin, the filamentous network resulting from a stepped up production or a subcellular stress fiber mobilization (Li et al., 2008; Pellegrin and Mellor, 2007), in reaction to the glyphosate treatment, could be the nature of the stiffness increase. Moreover, several studies can uphold this hypothesis: Birukova et al. (2009) have demonstrated the presence of an enhanced peripheral actin cytoskeleton and stiffness modifications after treatment with barrier-protective agents on endothelial cells; Chouinard et al. (2008) have inferred the correlation between actin cytoskeleton rearrangement and cell rigidity during ox-LDL-induced impairments on endothelial cells; finally, Rotsch and Radmacher (2000) showed that disaggregation of actin filaments results in decreasing elastic modulus of fibroblasts, while reorganization of microtubules did not affect cell elasticity. Thus, for low concentrations, it can be assumed that actin is overproduced in the case of cellular defense response against glyphosate.

However, in their review, Magin et al. (2007) have shown the involvement of keratin intermediate filaments in resistance towards multiple kinds of stresses. Furthermore, Lulevich et al. (2010) inferred that stiffness modifications in keratinocytes could principally be due to the keratin intermediate filament network.

Cytoskeleton structures are not visible any more after IC50 treatment but the cell membrane is impaired and cells are not so spread on the substrate. Statistical analysis confirms that cells are stiffer after a 15 mM-glyphosate treatment but YM values are comparable to the control after a 30 mM-exposition.

Since no differences, in terms of cytoskeletal network and mechanical parameters, were observed between control and IC50-glyphosate treated cells, we believe that an IC50 treatment is too drastic to allow cytoskeleton mobilization whereas a less cytotoxic treatment provokes a defense reaction in the cell. Impairments of the cytoskeleton and cytoplasmic membranes are required for cell shrinkage and loss of cell–cell contact to occur (Pellegrino et al., 2004). During glyphosate treatment, these events result in the cell detachment or fragmentation.

As previously shown in our laboratory (Gehin et al., 2005, 2006), glyphosate impairs cells via a significant loss of their antioxidant capacities. Quercetin is one of the most ubiquitous bioflavonoids in foods of plant origin. Although quercetin is generally considered for providing protection against oxidative injury and inflammation (Kook et al., 2008), as well as in keratinocytes (Kimura et al., 2009; Wang et al., 2010), recent studies have demonstrated that its cytoprotective effects occur within a narrow concentration range (Ossola et al., 2008; Shih et al., 2004). Additionally, quercetin inhibits the proliferation and migration of aortic smooth muscle cells (Alcocer et al., 2002). These conflicting effects were also noticed in our *in vitro* model (data not shown). The antioxidant treatment is based on the quercetin concentration of 100 mM which better prevents cell mortality during an 18 h-glyphosate treatment. At low glyphosate treatment (15 mM), quercetin provokes the disappearance of the cytoskeletal network with a “control-like” topography profile and a decrease of the cell stiffness. At IC50-glyphosate treatment, quercetin leads to the appearance of a cytoskeletal network and the decrease of the cell stiffness. It can be the result of a glyphosate effects reversion with a “15 mM-like” profile. When testing this classical antioxidant, it has been shown that the glyphosate-induced mechanical phenotype might be connected to the oxidative stress it created.

In conclusion, in this study, we were able to consistently demonstrate the impact of the glyphosate on cutaneous cellular mechanical properties as a consequence of oxidative stress. At low cytotoxicity, glyphosate enhanced the HaCaT cell stiffness and induced the emergence of cytoskeleton structures at subcellular scale. At high cytotoxicity conditions, HaCaT cells showed external morphological damages as a flattened external cell membrane and a reduced number of the protrusions, but no general

stiffness changes. Although, at low cytotoxicity, cytoskeleton mobilization is a defense mechanism, at high cytotoxicity, cells are overtaken by glyphosate-induced cellular events. Our study represents a step toward underlining a glyphosate-based *in vitro* cellular model of environmental epidermal aging.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.jsb.2012.02.007.

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